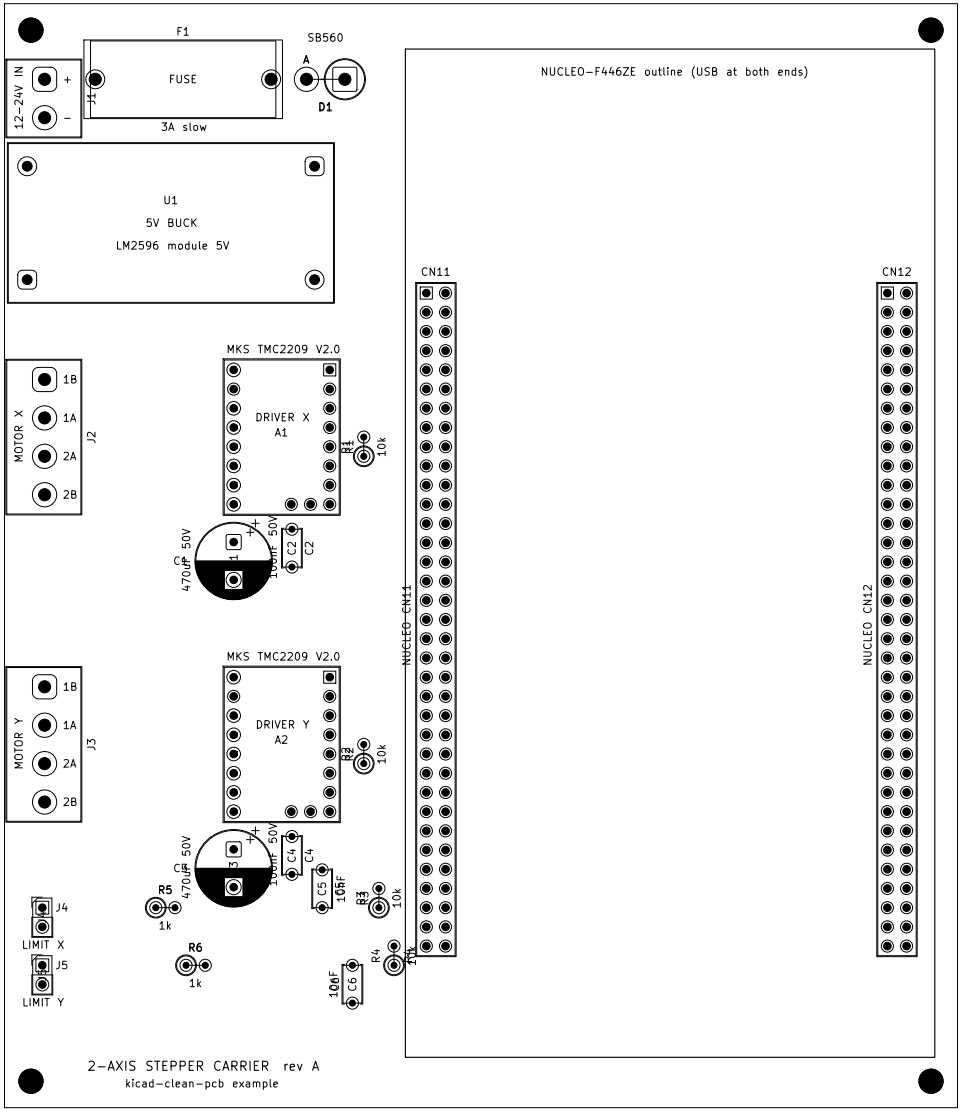

F1
SB560
12-24V IN
+
–
J1
FUSE
3A slow
A
D1
U1
5V BUCK
LM2596 module 5V
NUCLEO-F446ZE outline (USB at both ends)
CN11
CN12
NUCLEO_CN11
NUCLEO_CN12
MOTOR X
J2
1B
1A
2A
2B
MKS TMC2209 V2.0
DRIVER X
A1
R1
10k
C1
470uF 50V
C2
100nF 50V
MOTOR Y
J3
DRIVER Y
A2
R2
C3
C4
C5
10nF
R3
R4
R5
1k
R6
C6
10uF
LIMIT X
J4
LIMIT Y
J5
2-AXIS STEPPER CARRIER rev A
kicad-clean-pcb example